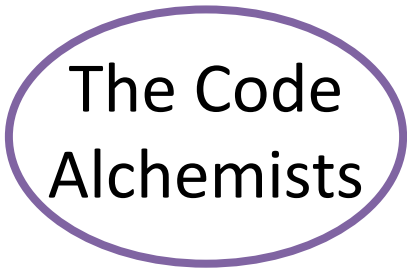


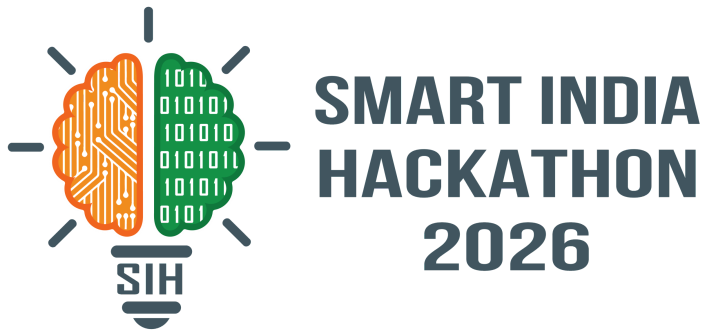
TITLE PAGE

- Problem Statement ID – SIH26178
- Problem Statement Title- A resilient, AI-powered environmental monitoring network that provides early detection, localized intelligence, and actionable alerts for floods, forest fires, pollution events, and other environmental hazards common in India, enabling authorities and communities to shift from reactive disaster response to proactive risk prevention.
- Theme- Disaster Management
- PS Category- Hardware
- Team ID- 165248
- Team Name – The Code Alchemists





OcuLA - Ocular Layer Architecture



PROBLEMS



DEATHS

1.67M

Annual Deaths from Air Pollution

Deaths in India attributable to air pollution each year — not including weather-related deaths, counted separately. Plus 2803 deaths from floods & storms in FY 2024–25.

Source: GBD 2019 / Lancet Planetary Health (2020); Ministry of Home Affairs.



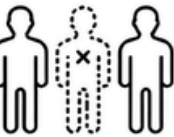
COST

562 vs ₹1.5–2 Cr

Real-Time Stations vs. Cost Per Station

India has only 562 real-time (continuous) air quality monitoring stations nationwide — each costing ₹1.5–2 crore to build.

Source: Centre for Science and Environment, State of India's Environment 2026; CPCB / State tender data.



COVERAGE

64%

Of Districts Have Zero Continuous Monitoring

Nearly two-thirds of India's ~742 districts — including 261 with over a million people — have no real-time monitoring at all. Even India's 966 manual stations are read only twice a week, not continuously.

Source: Centre for Science and Environment, State of India's Environment 2026; CPCB / NAMP network.



DELAY

13.5 HRS

Warning Issued, Never Acted On

A hyperlocal rain-gauge warning reached authorities 13 hours before the 2024 Wayanad landslide — it wasn't wired into the official alert system. 298 lives were lost.

Source: Onmanorama, Dec 2024; Hume Centre for Ecology and Wildlife Biology.



DATA INTEGRITY

Tampering Risk

Stations Can Be Manipulated

Delhi monitoring stations were reportedly sprayed with water or switched off to suppress readings during peak pollution, Oct–Dec 2025. CPCB disputes the allegations.

Source: The Straits Times / Times of India, Dec 2025; CPCB statement.

PROPOSED SOLUTION

- OcuLA is a distributed mesh network of **solar-powered nodes** that contains smart sensors.
- It uses **on-device AI** to spot natural hazards or predict upcoming disasters. It instantly **warns** remote communities of danger, keeping them safe even **when off-the-grid**.
- Ocula splits into **Child** and **Parent** Nodes to **maximize computational efficiency** and **minimize the overall cost**.

CHILD NODE: DETECT, ANALYSE & WARN

- Operating instantly and offline, this ESP32-based node handles continuous multi-sensor sampling.
- It executes local edge **AI inference**, and triggers instant local sirens and **alerts** during emergencies.

PARENT NODE: COLLECT, AUTHENTICATE & PREDICT

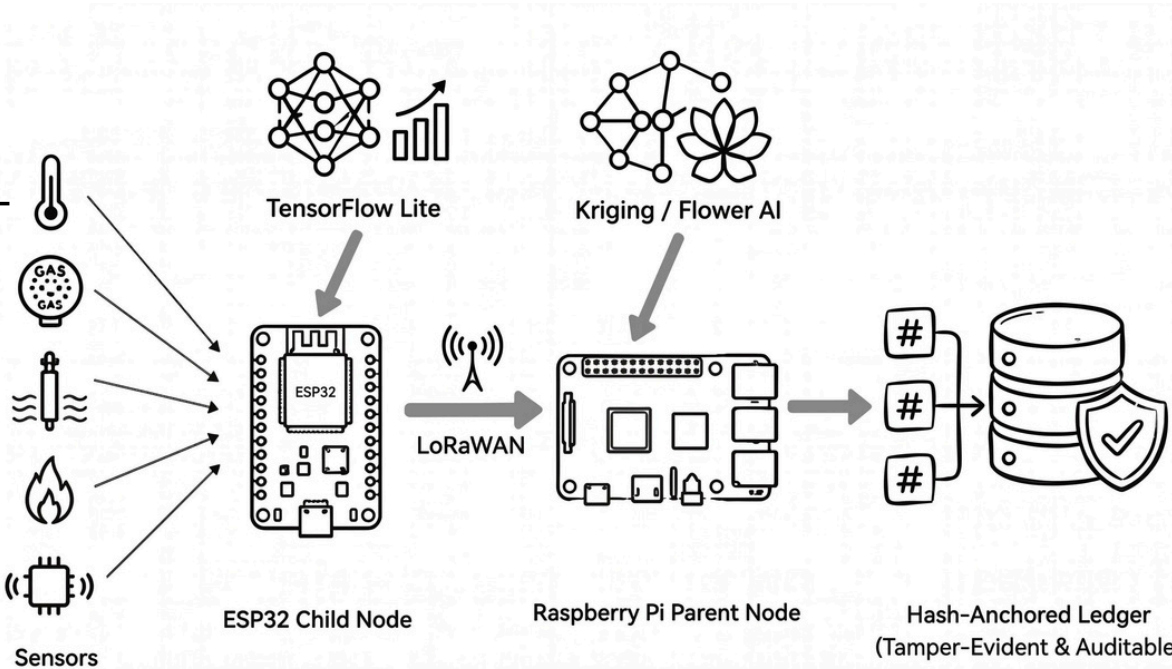
- A Raspberry Pi based node that possesses **hyperlocal awareness**, connected to **multiple** nearby child nodes to collect their data.
- Custom made linux ISO (OcuLA OS) with powerful on-device AI to cross-checks their **data authenticity** and **predict** upcoming disasters.

SOFTWARE & DATA INTEGRITY:

- A user friendly software dashboard designed to **manage nodes** and analyse **historical data**. --> [\[Live Dashboard\]](#)
- After authentication parent node pushes the data into a centralized, **hash-anchored ledger** to protect the **data integrity**.

INTRINSIC TRANSMISSION:

- Nodes form a resilient hierarchical star network using **LoRaWAN**.
- It gives **5 to 10km** range and can function even during a disaster.



Technology Stack ----- [GitHub Repo](#)

1. Edge Hardware (Child Node)

- **ESP32** microcontroller
- MQ-135 (gas/air quality), DHT11 (temp/humidity), ADXL345 (vibration/shock), flame sensor, water-level probe
- 128×64 OLED (SH1106G) + active buzzer for local alerts

2. Edge Intelligence (Child Node)

- **TensorFlow Lite Micro** — on-device inference, works without connectivity
- **CUSUM** statistical drift detection (gas), FFT-based vibration classification

3. Network Layer

- **LoRaWAN** (SX1276 module) — long-range, low-power child-to-parent uplink
- **ChirpStack** — open-source LoRaWAN network server on the parent node
- **Mosquitto (MQTT)** — internal message broker between ingestion and processing

4. Parent Node Hardware

- **Raspberry Pi 3b+**
- Solar panel + charge controller + battery for off-grid power

5. Aggregation & Forecasting (Parent Node)

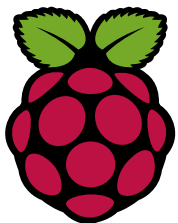
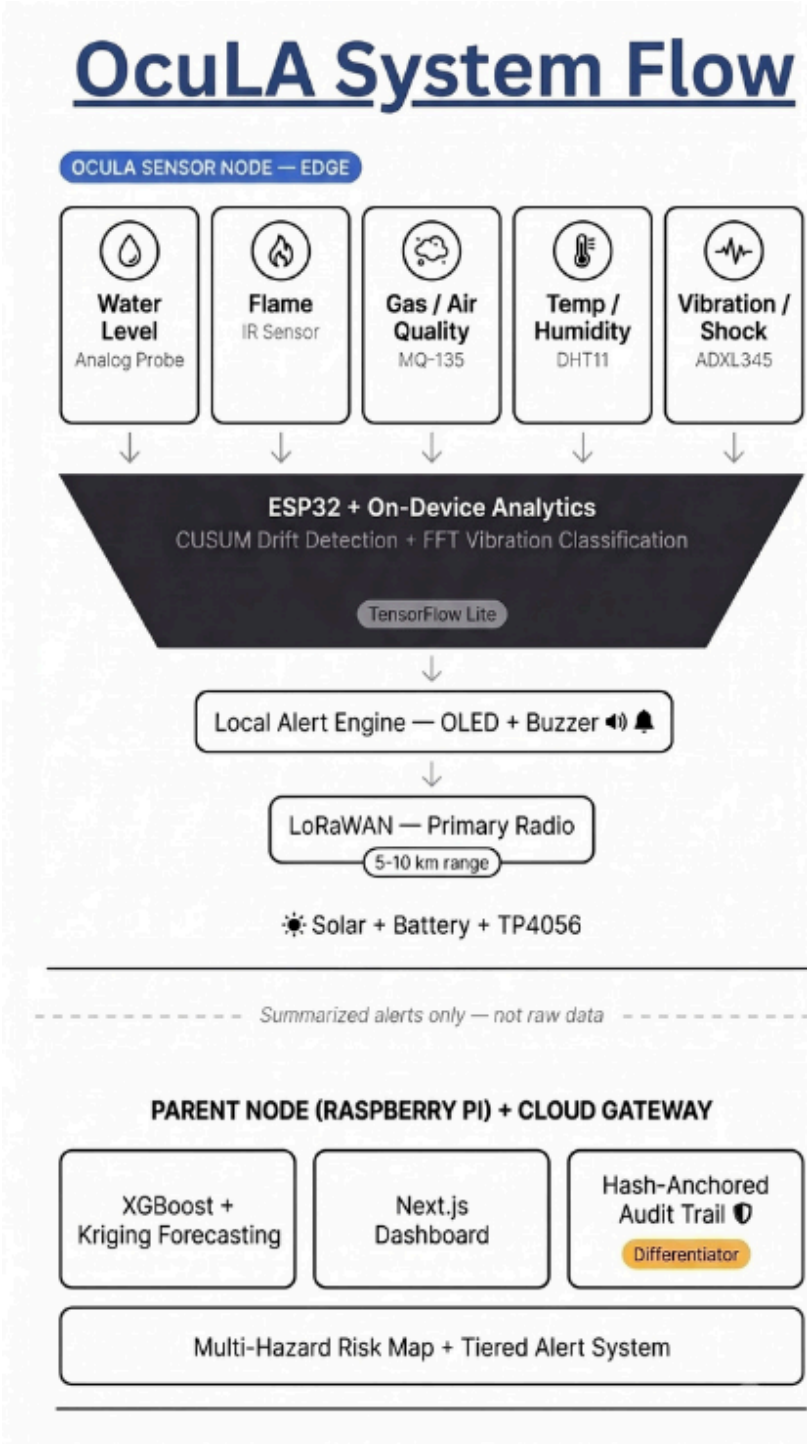
- **PyKrig** — spatial interpolation (Kriging) across the sensor mesh
- **Flower (flwr)** — federated learning aggregation across nodes
- **XGBoost** — hazard-risk forecasting on aggregated features
- SQLite / InfluxDB — local buffering and time-series storage

6. Data Integrity Layer

- Hash-chain + Merkle-anchored integrity layer — SHA-256 chained, auditable hazard records

7. Application Layer

- **FastAPI** — REST/WebSocket backend
- **Open Meteo API** - Satellite data
- **Next.js** — dashboard frontend
- **Mappls** — interactive geo-mapped hazard visualization

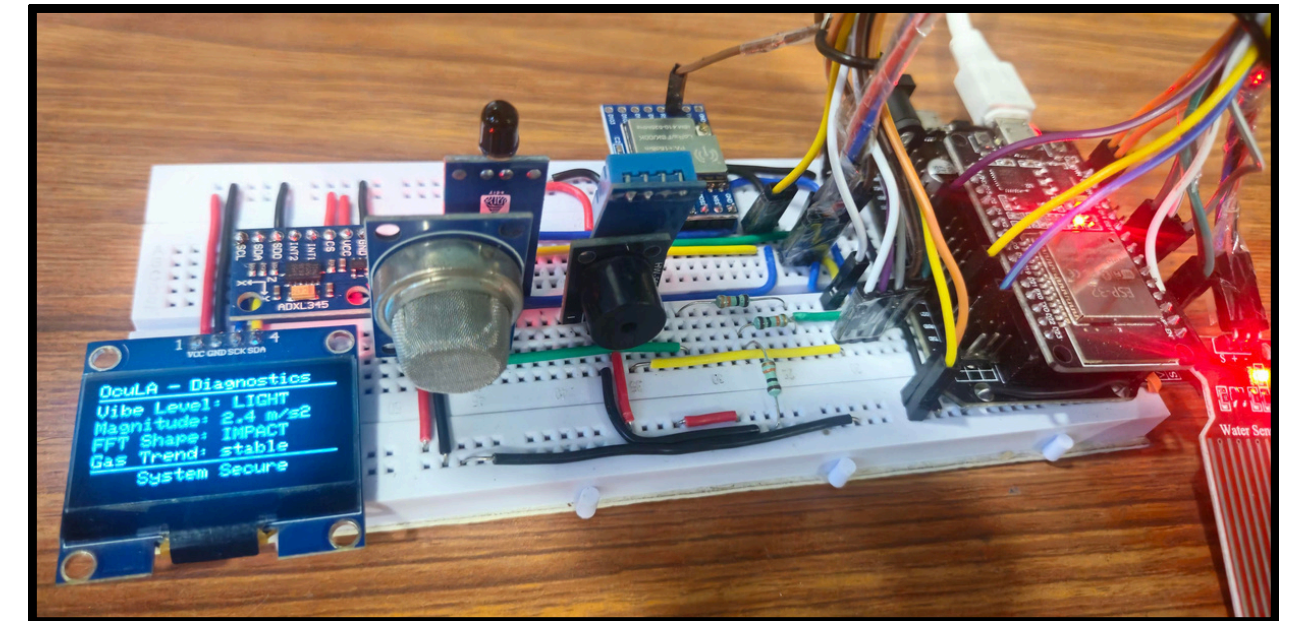


FEASIBILITY

- **Working Prototype:** This prototype features a fully functional hardware build utilizing the ESP32, combined with off-the-shelf sensors..
- **Radical Cost Reduction:** Each Child Node Costs ranges from ₹3,000 to ₹4,000, while the Parent Node is priced between ₹7,000 and ₹8,000.
- **Coverage:** Each node offers a range of 5 to 10 kilometers, enabling extensive area coverage with just a few nodes.

VIABILITY

- **Scalable Deployment Model:** Traditional station costs ₹1.5–2 Cr to covers 1 point. Meanwhile, OcuLA cluster (1 parent + 15 nodes) ≈ ₹60K that cover multiple points.
- $₹1.75 \text{ Cr} \div ₹60\text{K} \approx 291 \text{ clusters} \approx 4,360 \text{ sensors}$ for the price of 1 station (~4,375× cheaper per point)
- **Tamper-Evident Transparency:** Hash-anchored data provenance keeps historical data publicly visible, with any manipulation cryptographically detectable.



OcuLA Monitoring Testing Prototype

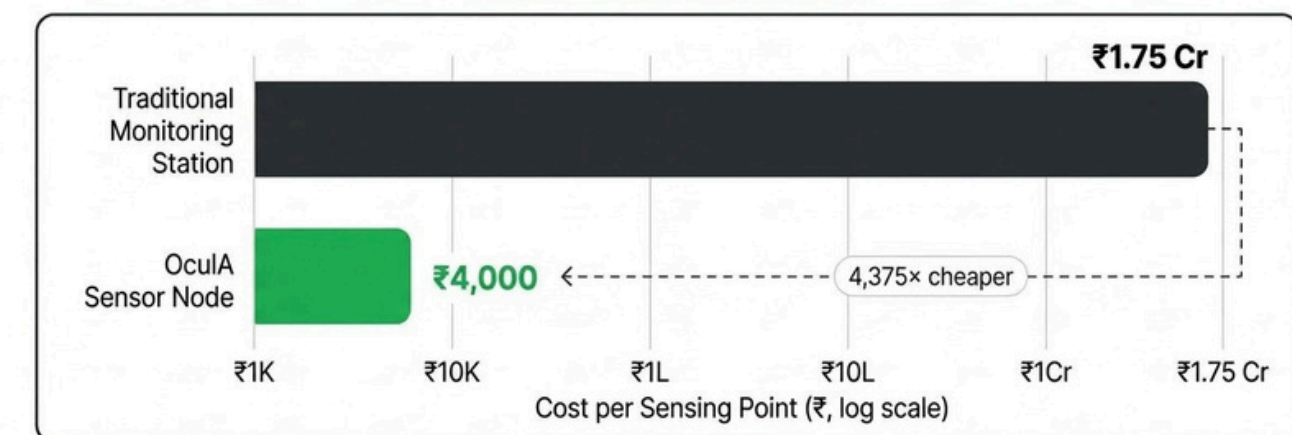
CHALLENGES & MITIGATIONS

- **LoRa dead zones:** overlap coverage so each node can reach more than one parent.
- **Sensor drift causing false alarms:** the CUSUM detector already tracks this drift statistically and self-calibrates against real noise data.
- **Connectivity gaps blocking the ledger:** local buzzer/OLED alerts fire instantly and independently; ledger writes just queue up and sync in a batch once connectivity returns.

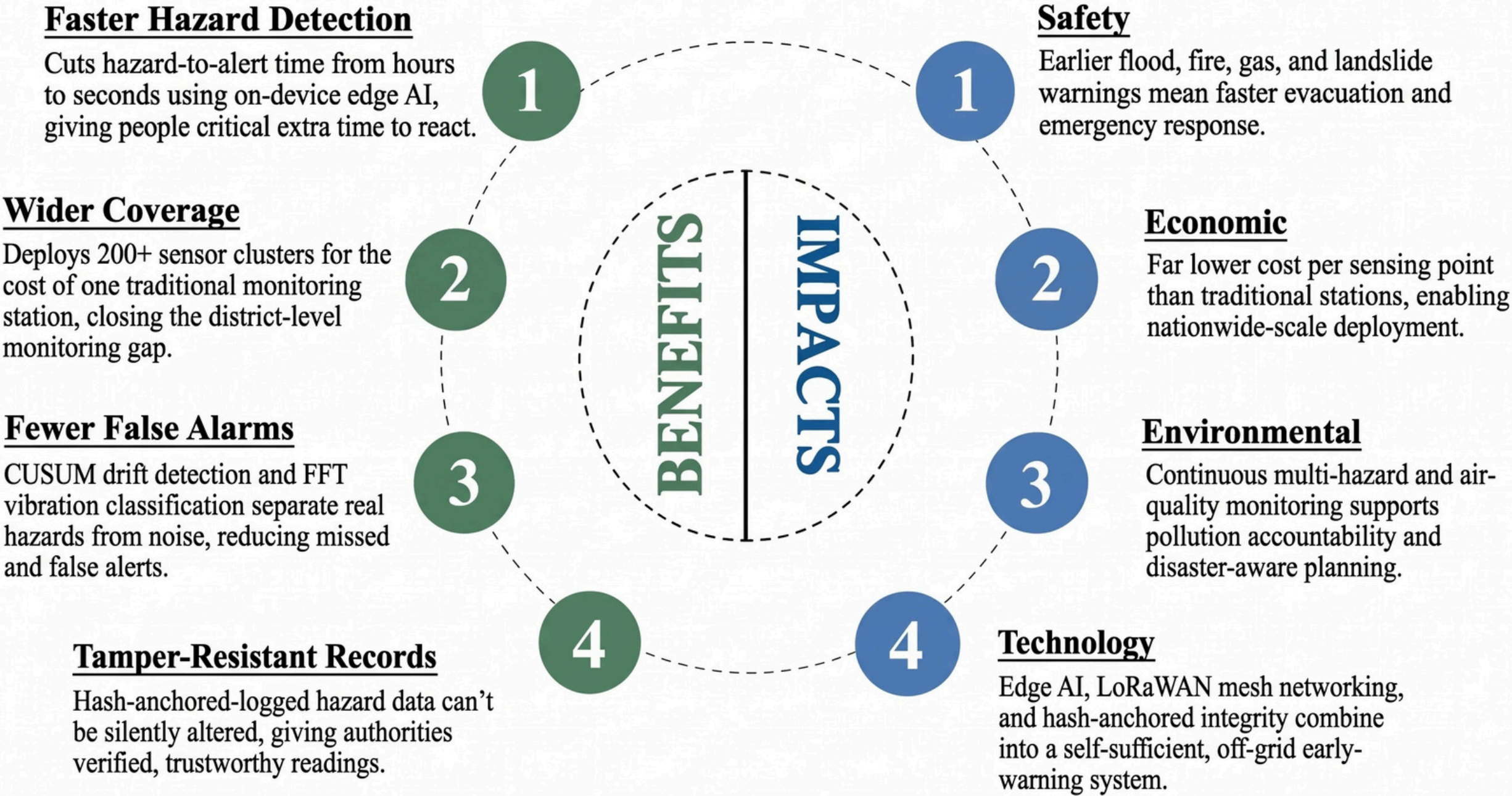
USE CASES

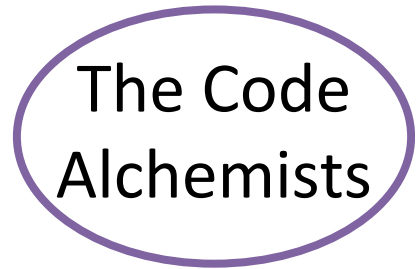
- **For Government:** Fills coverage gaps in existing monitoring networks at a fraction of the cost — reaching districts and regions current infrastructure doesn't.
- **For Research:** Provides hyperlocal environmental data that researchers can use across fields like climate science, disaster studies, and public health.
- **For Public:** Instant sirens and alerts give people real-time warning to take shelter before a hazard hits.

COST PER SENSING POINT: TRADITIONAL VS OCULA

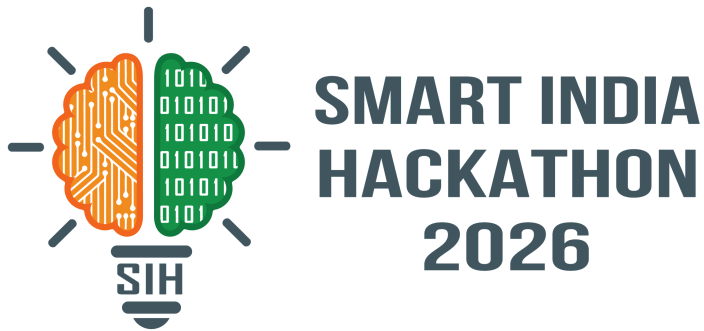


WAYANAD LANDSLIDE-2024 Could Ocula Have Made A Difference?





RESEARCH AND REFERENCES



MARKET RESEARCH

Core Statistics & Sources

- [Lancet/GBD 2019 – India Air Pollution Toll](#)
- [CSE – Air Quality Data Shadow Regions \(2025\)](#)
- [CPCB – NAMP](#)
- [WHO – Air Quality Guidelines](#)
- [MHA/Lok Sabha – Disaster Deaths FY24–25](#)
- [Pant et al. 2019 – PM Monitoring India](#)

Data Sources

- [OpenAQ](#)
- [World Air Quality Index](#)
- [Sentinel-5P \(TROPOMI\)](#)
- [NASA MODIS](#)
- [NASA VIIRS](#)

Case Studies

- [2024 Wayanad Landslides](#)
- [Wayanad Warning Timeline \(Onmanorama\)](#)
- [Wayanad Warning Timeline \(Mongabay\)](#)
- [Delhi AQI Tampering, 2025 \(Newslandry\)](#)
- [Delhi AQI Tampering, 2025 \(Business Standard\)](#)

Methodologies

- [Kriging & IDW Mapping, Delhi](#)
- [Random Forest PM2.5, Delhi](#)
- [RF + AOD Estimation, Maharashtra](#)
- [RF/XGBoost/DL Comparison, Jaipur](#)

Metric	OcuLA	Oizom	Aurassure	Airveda	Traditional CAAQMS
Cost / Node	₹4K	₹3L+	₹2L+	₹1.5L+	₹1.5–2 Cr
Offline Edge AI	✓ On-device (TFLite)	✗ Cloud-dependent	✗ Cloud-dependent	✗ Cloud-dependent	✗ Cloud-dependent
Network Backbone	✓ Native LoRaWAN mesh (5–10km)	📶 Cellular / Wi-Fi	📶 Cellular / Wi-Fi	📶 Cellular / Wi-Fi	📶 Wired / Fiber
Data Integrity	✓ Cryptographic hash-chain ledger	🛡️ Standard encryption	🛡️ Standard encryption	🛡️ Standard encryption	🗄️ Database logging
Off-Grid Solar	✓ Yes (Integrated/Optional)	✓ Yes (Integrated/Optional)	✓ Yes (Integrated/Optional)	✓ Yes (Integrated/Optional)	✗ Grid-only
Links	OcuLA	Oizom	Aurassure	Airveda	CAAQMS